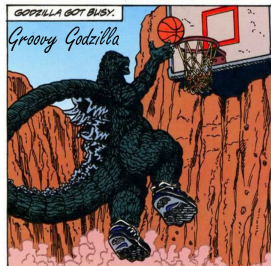


# Predicting March Madness Basketball Games

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# Introduction

- March Madness is a 68 team, single-elimination tournament held to determine the NCAA National Champion.
- The Kaggle competition (March Machine Learning Mania) evaluates the accuracy of predictions using the Brier score (BS).

$$BS = \frac{1}{N} \sum_{t=1}^N (f_t - o_t)^2 \quad (1)$$

- Where  $f_t$  is our prediction and  $o_t$  is the outcome.
- $o_t$  is 1 if the team wins, and 0 if they lose.

# Introduction

- A naive 'coin flip' strategy (50% probability for all games) yields a Brier Score of 0.25, serving as our baseline.

## Example

If we predict that Michigan has a win probability of 0.70 against Uconn.

If Duke **wins**,

$$\text{Cost} = (1 - 0.70)^2 = 0.09$$

If Duke **loses**,

$$\text{Cost} = (0 - 0.70)^2 = 0.49$$